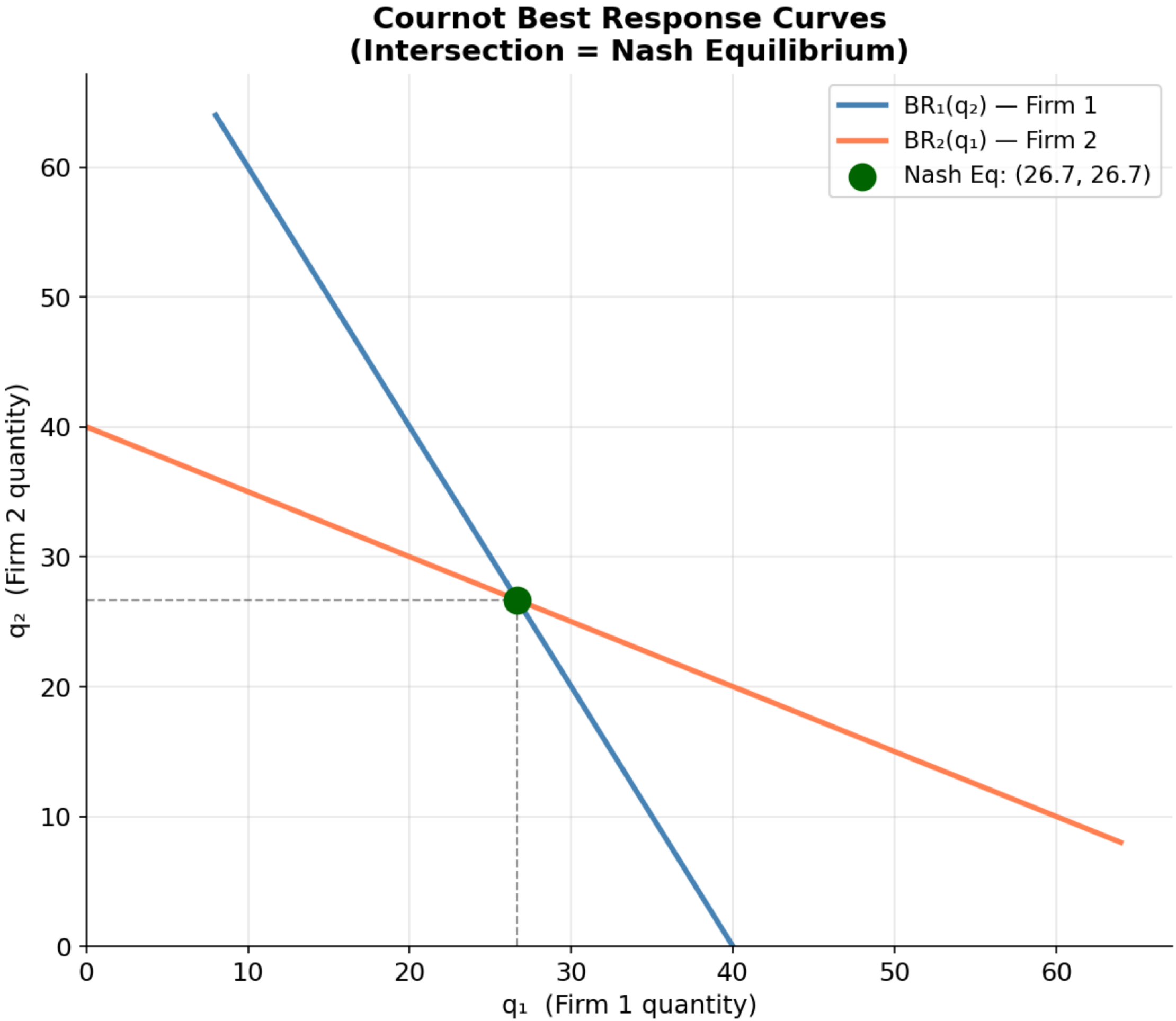
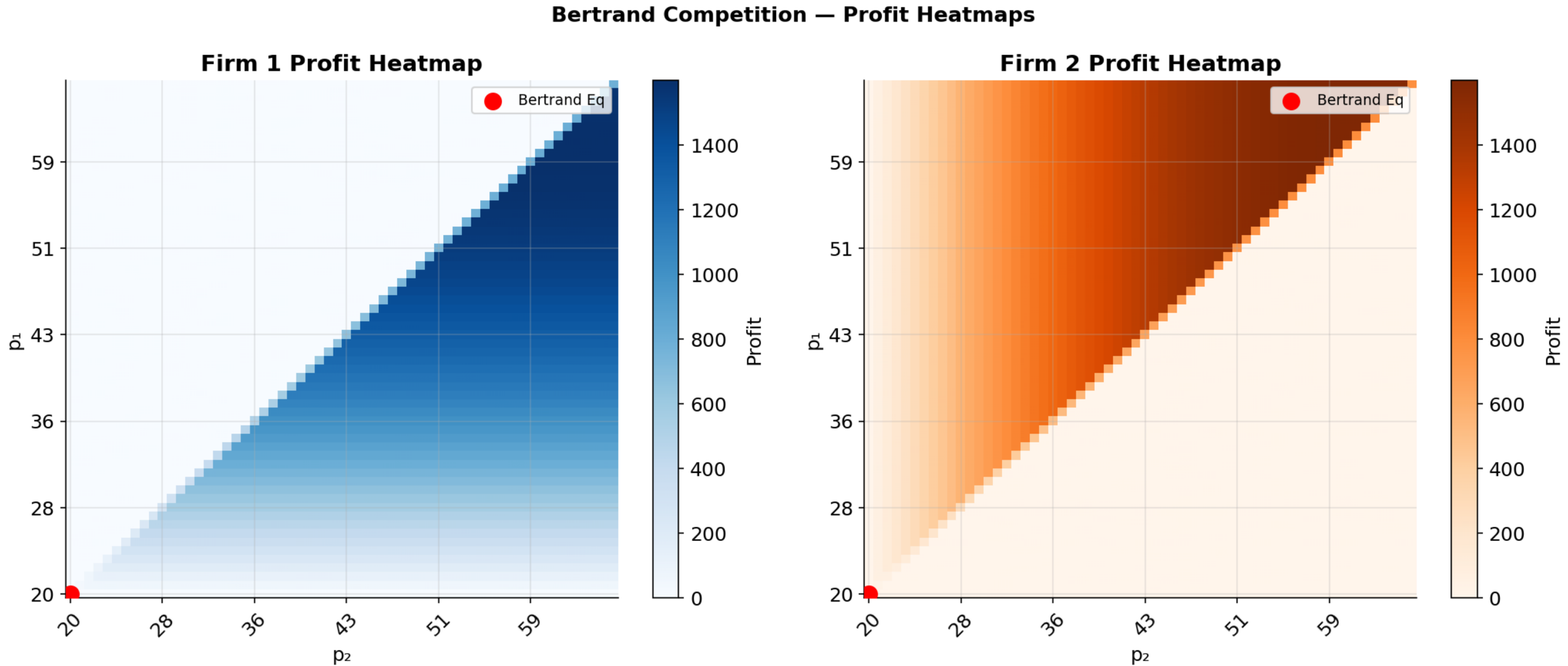
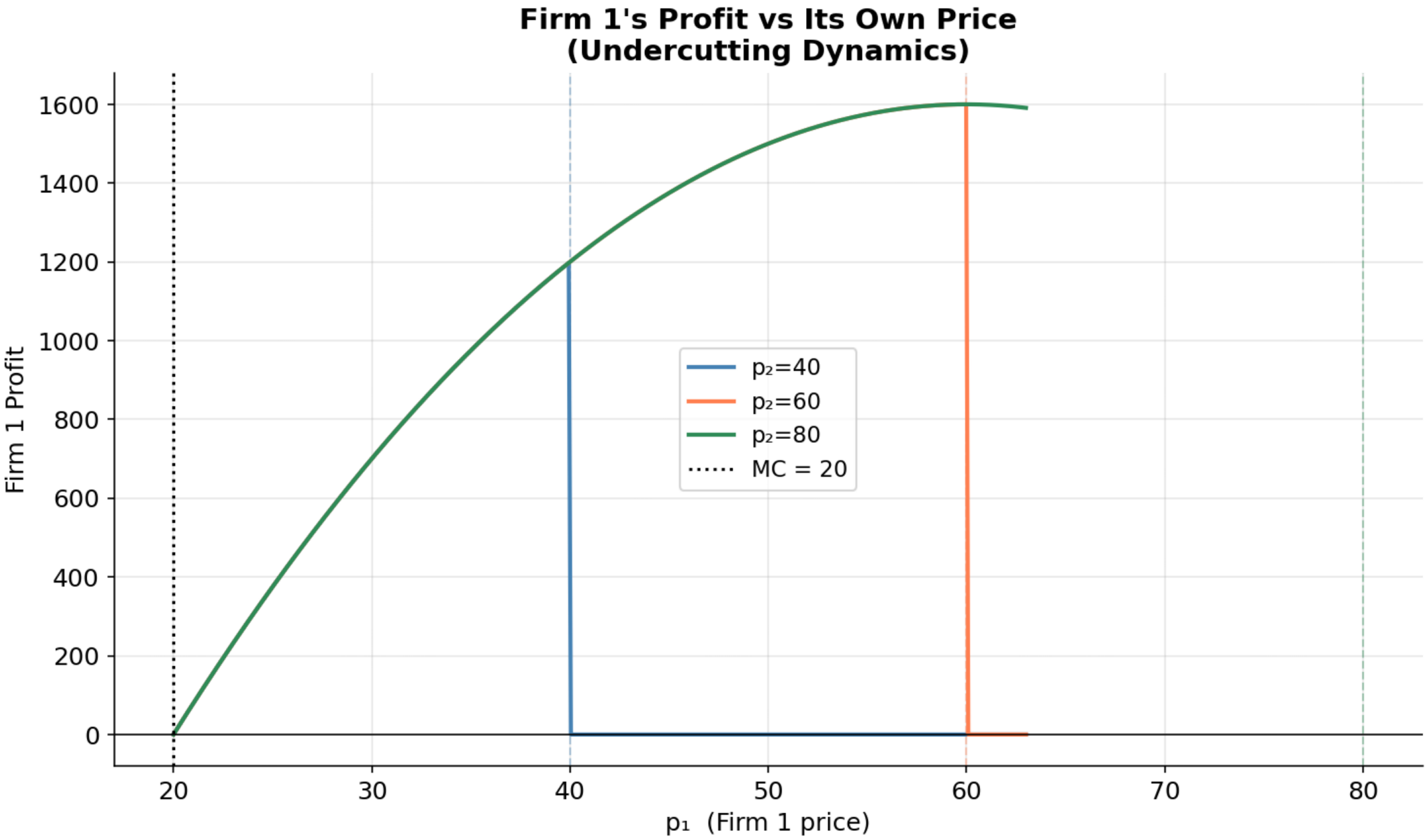
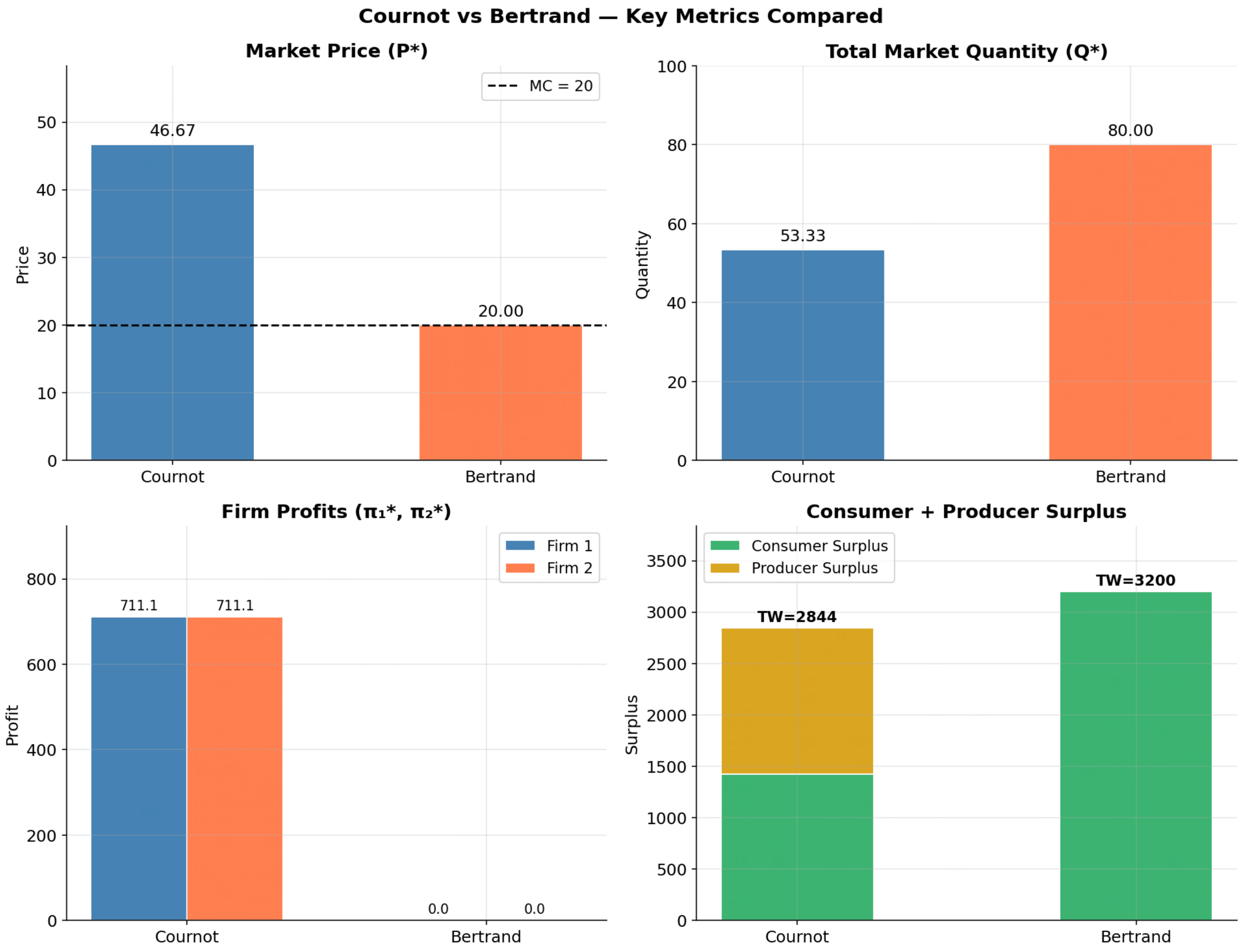
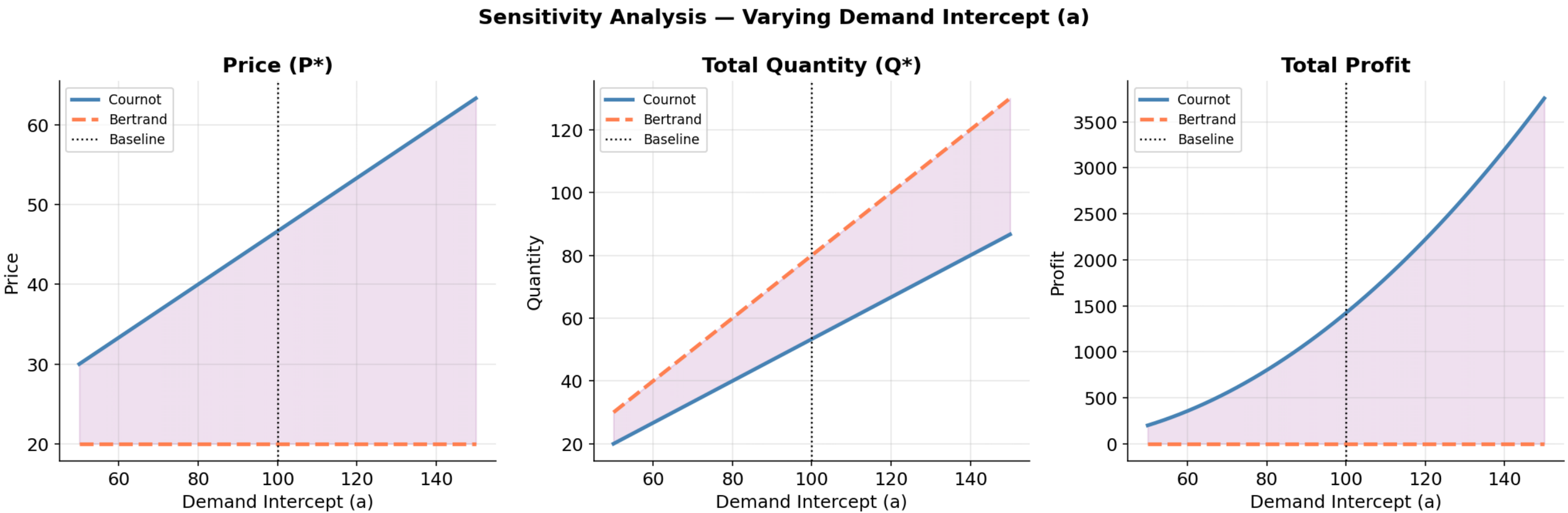
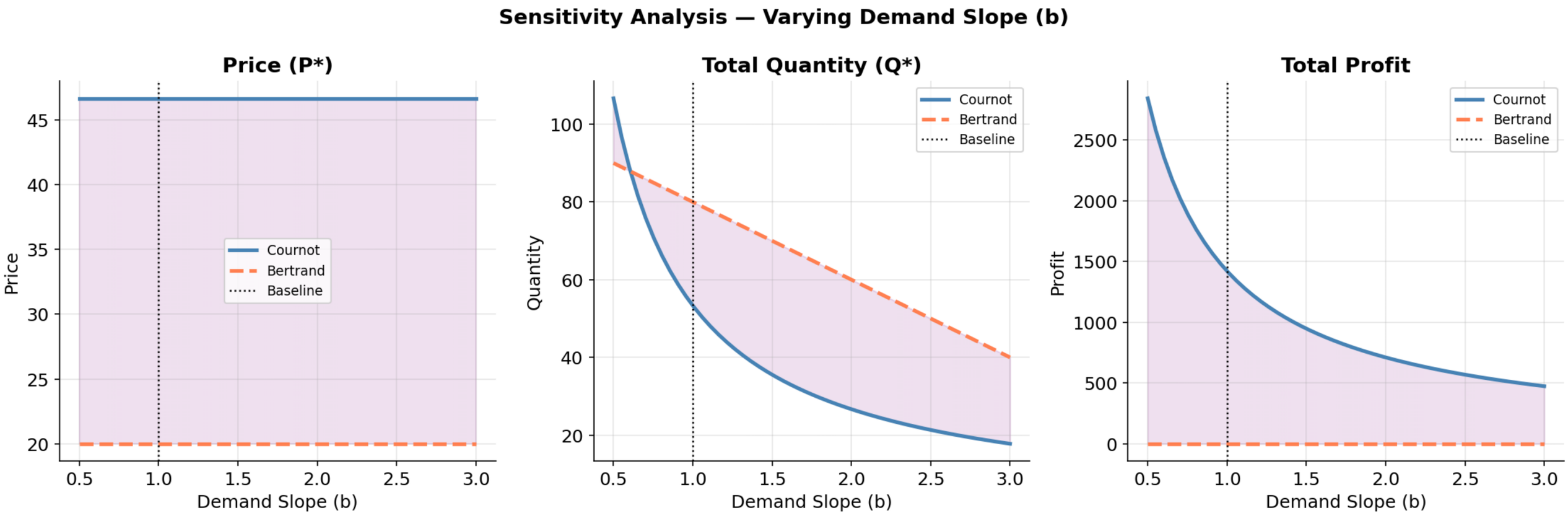

Cournot Best Response Curves
(Intersection = Nash Equilibrium)
BR₁(q₂) — Firm 1
BR₂(q₁) — Firm 2
Nash Eq: (26.7, 26.7)
q₂ (Firm 2 quantity)
q₁ (Firm 1 quantity)
0
10
20
30
40
50
60

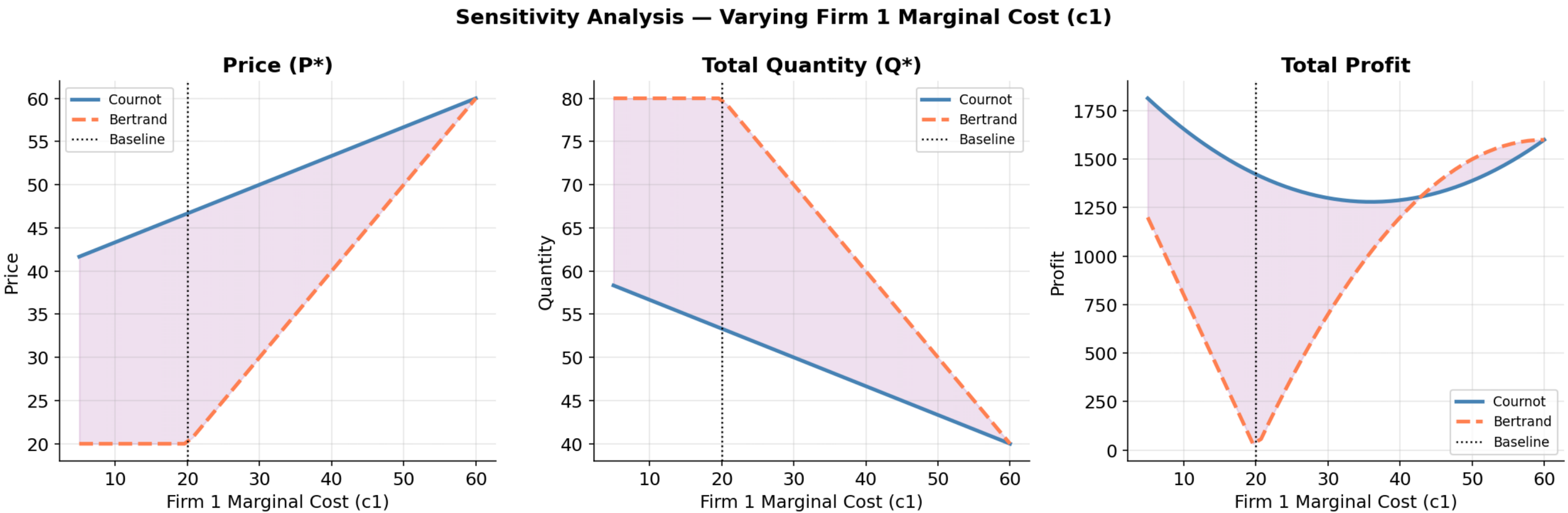

Bertrand Competition — Profit Heatmaps
Firm 1 Profit Heatmap
Bertrand Eq
p₁
p₂
20
28
36
43
51
59
Profit
0
200
400
600
800
1000
1200
1400
Firm 2 Profit Heatmap
Bertrand Eq
p₁
p₂
20
28
36
43
51
59
Profit
0
200
400
600
800
1000
1200
1400

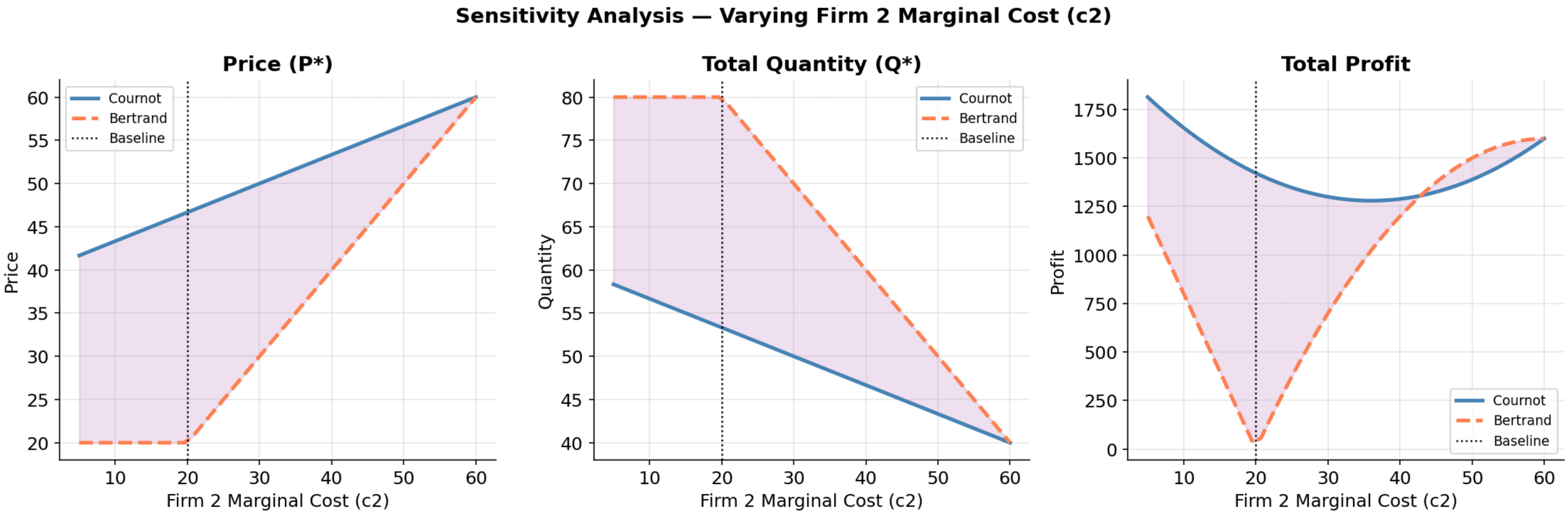

Firm 1's Profit vs Its Own Price
(Undercutting Dynamics)
Firm 1 Profit
1600
1400
1200
1000
800
600
400
200
0
20
30
40
50
60
70
80
p₁ (Firm 1 price)
p₂=40
p₂=60
p₂=80
MC = 20

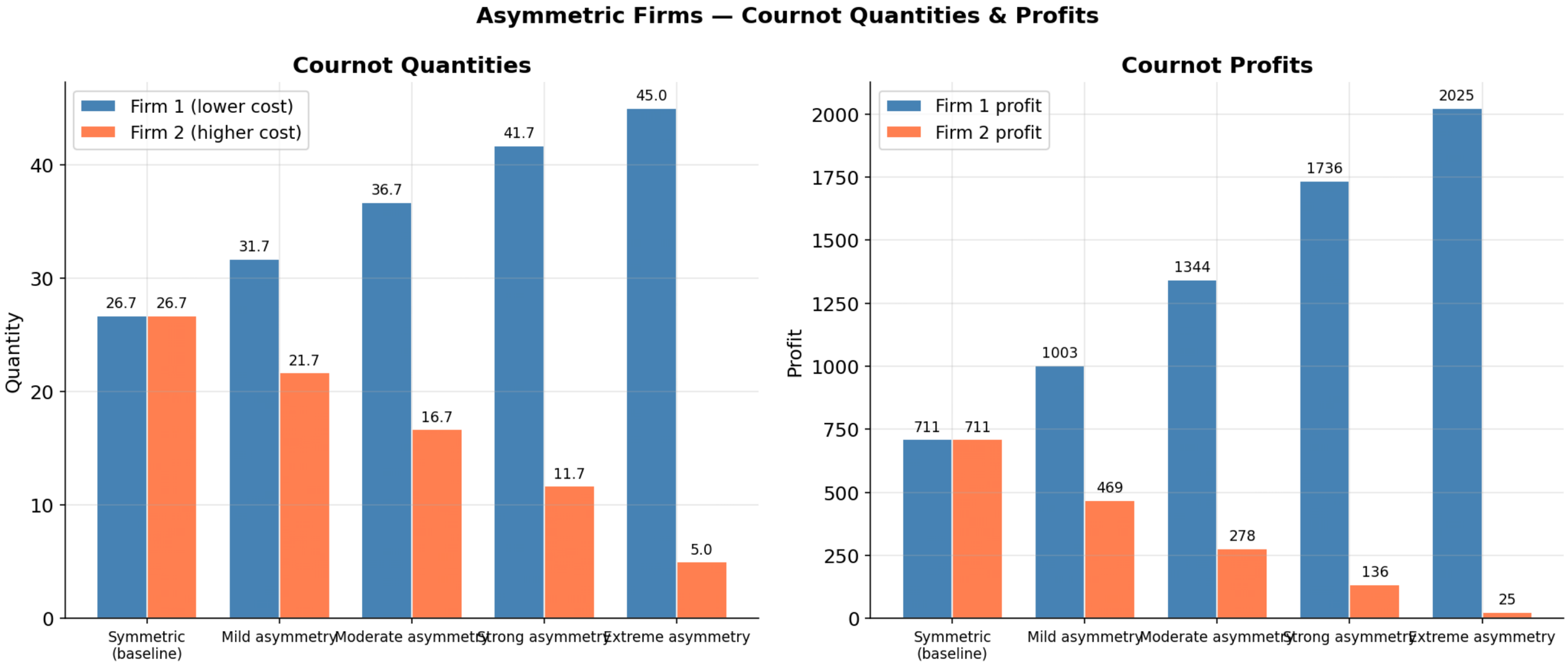
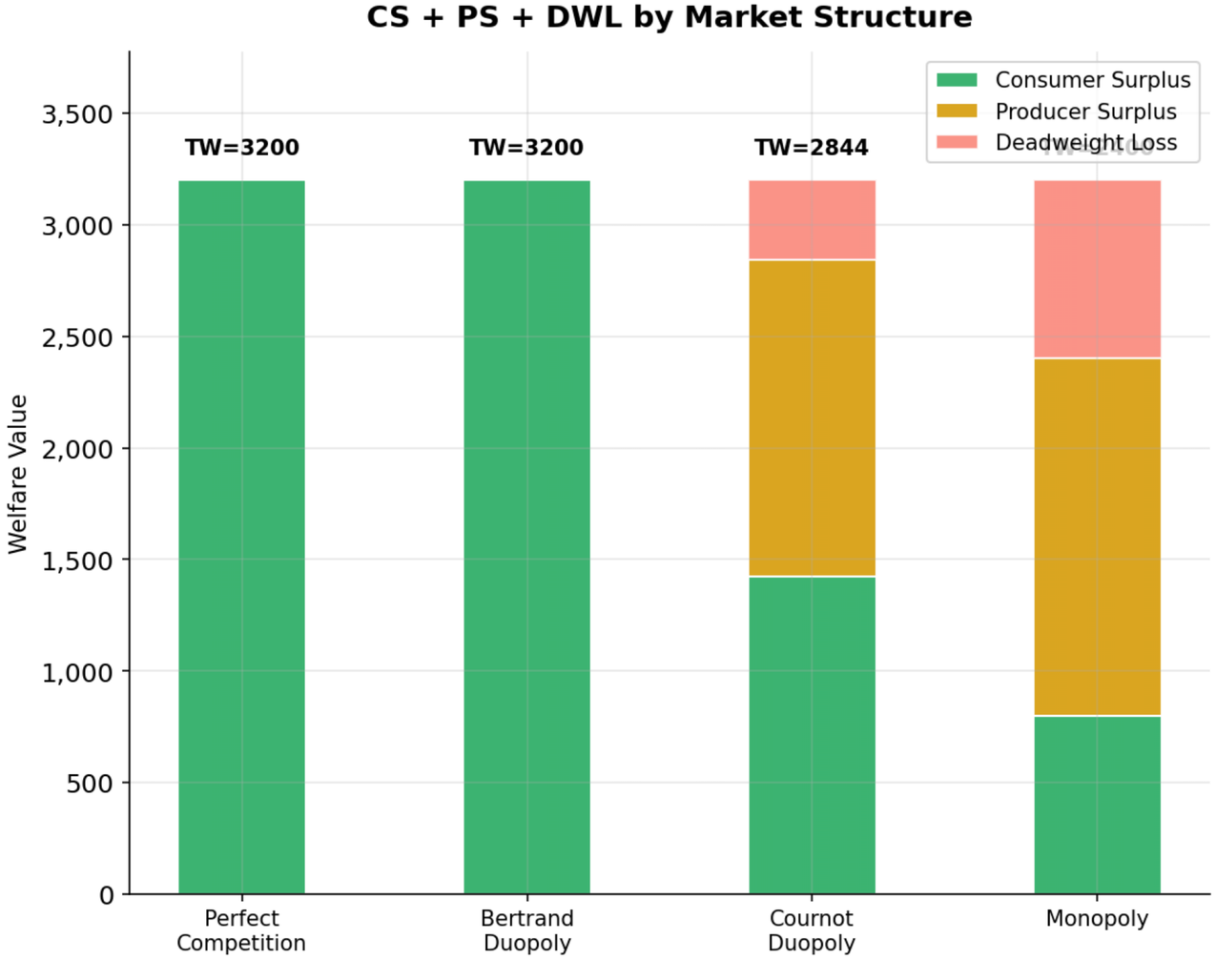
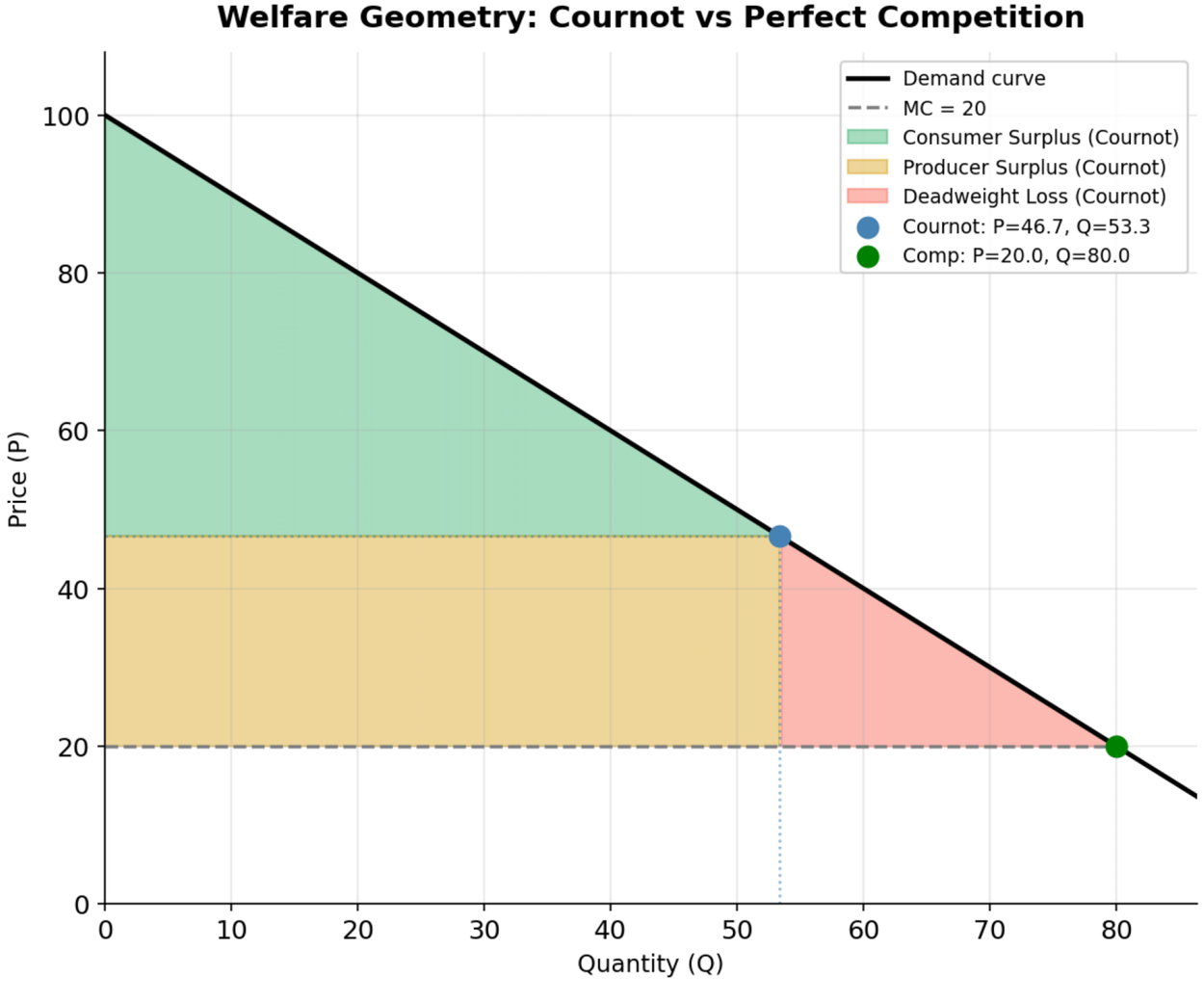

Cournot vs Bertrand — Key Metrics Compared
Market Price (P*)
MC = 20
Price
46.67
20.00
Cournot
Bertrand
Total Market Quantity (Q*)
Quantity
53.33
80.00
Cournot
Bertrand
Firm Profits (π₁*, π₂*)
Firm 1
Firm 2
Profit
711.1
711.1
0.0
0.0
Cournot
Bertrand
Consumer + Producer Surplus
Consumer Surplus
Producer Surplus
Surplus
TW=2844
TW=3200
Cournot
Bertrand


Sensitivity Analysis — Varying Demand Intercept (a)
Price (P*)
Total Quantity (Q*)
Total Profit
Cournot
Bertrand
Baseline
Price
Quantity
Profit
Demand Intercept (a)
20
30
40
50
60
80
100
120
140
0
500
1000
1500
2000
2500
3000
3500


Sensitivity Analysis — Varying Demand Slope (b)
Price (P*)
Price
Demand Slope (b)
Cournot
Bertrand
Baseline
Total Quantity (Q*)
Quantity
Demand Slope (b)
Cournot
Bertrand
Baseline
Total Profit
Profit
Demand Slope (b)
Cournot
Bertrand
Baseline


Sensitivity Analysis — Varying Firm 1 Marginal Cost (c1)
Price (P*)
Total Quantity (Q*)
Total Profit
Cournot
Bertrand
Baseline
Price
Quantity
Profit
Firm 1 Marginal Cost (c1)
Firm 1 Marginal Cost (c1)
Firm 1 Marginal Cost (c1)


Sensitivity Analysis — Varying Firm 2 Marginal Cost (c2)
Price (P*)
Total Quantity (Q*)
Total Profit
Price
Quantity
Profit
Firm 2 Marginal Cost (c2)
Cournot
Bertrand
Baseline


Asymmetric Firms — Cournot Quantities & Profits
Cournot Quantities
Firm 1 (lower cost)
Firm 2 (higher cost)
Quantity
0
10
20
30
40
26.7
26.7
31.7
21.7
36.7
16.7
41.7
11.7
45.0
5.0
Symmetric (baseline)
Mild asymmetry
Moderate asymmetry
Strong asymmetry
Extreme asymmetry
Cournot Profits
Firm 1 profit
Firm 2 profit
Profit
0
250
500
750
1000
1250
1500
1750
2000
711
711
1003
469
1344
278
1736
136
2025
25
Symmetric (baseline)
Mild asymmetry
Moderate asymmetry
Strong asymmetry
Extreme asymmetry

Extension 4: Welfare Analysis Across Market Structures